



Curriculum Vitae

Salah Faisal Saeed Saeed

📍 Sina'a, Taiz, Yemen

☎ (+967) 777 612 071 - 738 063 163 – 774 093 931

✉ salahfaisal589@gmail.com

🌐 <https://www.linkedin.com/in/eng-salah-faisal-saeed-saeed-naser-925a1b2a3>

Sex Male | Date of birth 12/11/1998 | Nationality Yemeni

PERSONAL STATEMENT

Mechatronics and Robotics Engineer (B.Sc., 2025) with an applied and research-oriented profile at the intersection of computer science, robotics, embedded systems, artificial intelligence, computer vision, MATLAB/Simulink-based modeling, control systems, optimization algorithms, circuit simulation, and laboratory-based technical education. I also hold a Vocational Professional Diploma in Industrial Control Systems/Electronics, completed with 92.27% (Excellent), ranked first in my cohort and second nationally in Yemen.

My experience combines hands-on engineering implementation, programming instruction, laboratory development, and academic research mentorship. I have delivered training in Python, C++, MATLAB/Simulink, embedded systems, IoT, circuit simulation, and optimization algorithms, while mentoring students in research methodology, literature review, academic writing, citation practices, manuscript structuring, and scholarly revision. I have contributed to published IEEE conference papers in robotics, robotic manipulators, computer vision, camera-based measurement systems, and tDCS, as well as accepted eSmarTA 2026 manuscripts in EV fast charging, EV charging systems, federated learning for healthcare, and deep-learning-based imaging through scattering media.

My current research work focuses on UAV control, PV-integrated bidirectional EV fast charging, off-grid hybrid renewable energy system optimization, federated learning, cybersecurity, security digital twins, and AI-assisted technical analysis. I seek to contribute to a Computer Science academic environment through teaching support, laboratory instruction, student mentoring, applied research assistance, and practice-oriented learning that connects programming, intelligent systems, robotics, and real-world problem solving.

WORK EXPERIENCE

08/2025 – 04/2026 **Scientific Research & Academic Writing Trainer — Student Research Mentorship**
Al-Shaheeda Na'mah Rassam School, Taiz, Yemen

- Led a nine-month Scientific Research and Academic Writing Training Program for female students from August 2025 to April 2026, focusing on research methodology, literature search, critical reading, academic writing, citation practices, manuscript structuring, academic integrity, and scholarly revision.
- Supervised student involvement in accepted research papers for the 6th International Conference of Smart Technology and Applications (eSmarTA 2026), supporting literature review, structured summaries, citation checking, manuscript refinement, and revision preparation.
- Contributed to strengthening the school's STEM research culture by translating advanced academic research practices into an accessible and structured learning experience for school-level students.

2024–Present **Scientific Research & Academic Writing Trainer — Student Research Mentorship**
Ne'marassam School, Taiz, Yemen

- Mentored students in research methodology, literature review practice, academic writing, citation discipline, and publication-oriented manuscript preparation.
- Guided students in formulating research questions, organizing technical arguments, preparing structured manuscripts, and revising work according to academic feedback.
- Supported the development of research communication skills relevant to engineering, STEM education, and conference-style presentation.

2024 **Optimization Algorithms Trainer (ABC) — MATLAB Integration Program**
Ne'marassam School, Taiz, Yemen

- Delivered a 21-day applied training program on the Artificial Bee Colony algorithm, objective-function formulation, parameter tuning, and convergence interpretation.
- Implemented MATLAB-based optimization workflows to support simulation-driven evaluation, baseline comparison, and analytical interpretation of engineering results.

2024 Embedded Systems & IoT Instructor — Student Training Program
Ne'marassam School, Taiz, Yemen

- Delivered hands-on training in embedded systems and IoT, covering microcontroller fundamentals, sensor/actuator interfacing, and prototype development.
- Supervised student activities using Arduino, ESP32, and Raspberry Pi, with emphasis on testing, debugging, connectivity workflows, and technical documentation.
- Strengthened students' applied engineering skills in hardware–software integration, monitoring/control logic, and practical problem solving.

2023 MATLAB & Simulink Trainer — Student Training Program
Ne'marassam School, Taiz, Yemen

- Delivered a two-month training program in MATLAB programming and Simulink-based modeling for engineering applications.
- Trained students in numerical computing, data visualization, model construction, simulation configuration, debugging, and output interpretation.
- Guided applied mini-projects that connected theoretical concepts with simulation-based implementation and structured verification.

2023 Programming Instructor (Python & C++) — Student Training Program
Ne'marassam School, Taiz, Yemen

- Delivered a two-month training program in Python and C++ programming fundamentals for STEM-oriented applications.
- Trained students in algorithmic thinking, modular code organization, debugging practice, and foundational object-oriented programming concepts.
- Linked programming exercises to robotics, embedded systems, and microcontroller-related workflows to strengthen applied technical problem solving.

2023 Circuit Simulation Trainer (Multisim & Proteus) — Student Training Program
Ne'marassam School, Taiz, Yemen

- Delivered a one-month training program in analog and digital circuit simulation using Multisim and Proteus.
- Trained students in schematic verification, virtual measurement, circuit validation, and systematic troubleshooting.
- Designed simulation-based tasks linking circuit theory with practical electronics and embedded-systems readiness.

2023 Robotics Laboratory Development & STEM Technical Training Specialist
Ne'marassam School, Taiz, Yemen

- Contributed to establishing and developing the school's robotics laboratory, including technical organization, equipment readiness, prototyping workflows, and hands-on STEM learning activities.
- Supported the integration of microcontroller platforms, sensors, actuators, embedded-system tools, and educational robotics components for practical student projects.
- Prepared laboratory documentation, technical checklists, basic maintenance guidance, and structured procedures to improve safety, reliability, and operational readiness.
- Delivered technical training in applied electronics, Multisim and Proteus, MATLAB/Simulink, Python/C++, Embedded Systems and IoT, and ABC optimization using MATLAB-based engineering models.
- Supervised electronics and measurement activities involving circuit testing, measurement accuracy, troubleshooting, and laboratory safety.
- Mentored students in research methodology, academic writing, and publication-oriented revision, supporting student work published in IEEE Xplore-indexed papers.

- 6-Month Period **Industrial Training Assistant — Industrial Control Systems**
Technical Industrial Institute, Taiz, Yemen
- Assisted in practical training activities related to industrial control systems and electronics.
 - Supported hands-on instruction in control circuits, electronic components, panel-level practice, and troubleshooting.
 - Contributed to vocational laboratory sessions and strengthened early exposure to applied industrial-control training.
- 2023–2024 **Computer Maintenance Engineer & Physics Laboratory Technician**
AL-ASBAHI Private School, Taiz, Yemen
- Performed preventive and corrective maintenance of computer systems, operating environments, educational software, and basic network connectivity.
 - Prepared physics laboratory experiments, organized equipment, supported practical sessions, and maintained safe experimental procedures.
 - Provided technical support to staff and students while improving the reliability of classroom and laboratory infrastructure.
- 2022–Present **Physics Laboratory Engineer & Technical Specialist**
Ne'marassam School, Taiz, Yemen
- Maintained laboratory readiness through equipment inspection, baseline maintenance, experiment setup refinement, and safety-aware operating procedures.
 - Provided technical support during practical sessions and improved documentation practices to strengthen reliability, repeatability, and scientific rigor.
 - Assisted teachers and students in using laboratory instruments responsibly while supporting applied STEM learning outcomes.
- 2022–Present **Electronics & Measurement Instruments Instructor**
Ne'marassam School, Taiz, Yemen
- Delivered structured instruction combining electronics theory, measurement instrumentation, and hands-on laboratory practice.
 - Supervised practical activities involving measurement accuracy, circuit testing, fault diagnosis, safety practice, and systematic troubleshooting.
 - Developed applied exercises and mini-project guidance to strengthen students' technical reasoning and engineering curiosity.
- 2022 **Applied Electronics Trainer — Physics Teachers Professional Development**
Ne'marassam School, Taiz, Yemen
- Designed and delivered a five-month applied-electronics program for physics teachers covering components, instrumentation, analog/digital basics, microcontrollers, PCB prototyping, and soldering practice.
 - Integrated troubleshooting exercises and introductory power-electronics concepts to support stronger laboratory instruction and practical demonstration.
- 2022 **Circuit Simulation Trainer (Multisim & Proteus) — Physics Teachers Program**
Ne'marassam School, Taiz, Yemen
- Trained teachers in simulation-based circuit design, virtual measurement, debugging, and preparation of curriculum-aligned laboratory activities.
 - Supported the use of Multisim and Proteus as instructional tools for bridging theoretical concepts with practical electronics learning.
- 2022–2023 **Physics Teacher — Secondary Level**
Al-Ghad Al-Mushreq School, Taiz, Yemen
- Delivered secondary-level physics instruction integrating theoretical explanation, structured problem solving, and practical demonstration.
 - Supported laboratory-based learning and maintained an organized, disciplined, and student-centered classroom environment.
- 2023–2024 **Student Training Specialist — Robotics, Embedded Systems & Applied Electronics**
Ne'marassam School, Taiz, Yemen
- Delivered structured hands-on training in robotics, embedded systems, programming, and applied electronics.

- Supervised student prototyping activities involving hardware–software integration, circuit testing, debugging, and technical documentation.
- Strengthened students' practical engineering competence and readiness for advanced STEM and technical projects.

PORTFOLIO AND SUPPORTING EVIDENCE

[Selected documentary and visual evidence of my teaching, technical training, laboratory work, student research mentorship, graduation-project activities, ongoing research, and applied maintenance work is available through the following links:](#)

Selected Institutional Announcements and Professional Evidence:

<https://drive.google.com/drive/folders/1s36oismPFgLLrYATrjibdvow2JUvmfc?usp=sharing>

Teaching, Research, and Technical Portfolio:

https://drive.google.com/drive/folders/1hsCH4zNxgabUPjg6cfnh_wXnQRx6CL1K?usp=sharing

EDUCATION

- 07/2025 **B.Sc. in Engineering (Mechatronics and Robotics Engineering)**
Taiz University, Alsaeed Faculty for Engineering & Information Technology, Taiz, Yemen
 Cumulative grade: 76.54% (Good)
 Graduation project: Manipulator Robot Error Correction Using Computer Vision
- 2016/2017 **Vocational Professional Diploma (Industrial Control Systems / Electronics)**
Technical Industrial Institute (Baghdad.st), Sana'a, Yemen
 Cumulative grade: 92.27% (Excellent). Ranked Second Republic.

JOB-RELATED SKILLS

- **Programming:** Python, C/C++, MATLAB; disciplined debugging and documentation.
- **Modeling & Simulation:** MATLAB/Simulink; simulation-driven evaluation workflows.
- **Robotics Middleware & Simulation:** ROS 2; nodes/topics, workspace setup, and Gazebo-based simulation workflows.
- **SolidWorks & Mechanical Design:** SolidWorks; 3D modeling, technical drawings, and assembly-level design for mechatronics prototypes.
- **Circuit Simulation & PCB:** Proteus; schematic capture, simulation-based verification, and PCB workflow fundamentals.
- **Electronics & Instrumentation:** circuit analysis, measurement instruments, fault diagnosis and troubleshooting.
- **Robotics & Embedded Systems:** microcontroller-based prototyping, sensors/actuators integration, laboratory workflow setup.
- **IoT:** connectivity-oriented prototyping and monitoring/control concepts (educational implementations).
- **Optimization:** metaheuristic optimization (ABC) and MATLAB model integration.
- **Research & Writing:** literature synthesis, technical writing, manuscript preparation, and presentation delivery.
- **Scientific Typesetting (LaTeX):** professional document preparation for academic manuscripts, tables/figures, and references.
- **Productivity Tools:** Microsoft Office (Word, PowerPoint, Excel) — professional reporting, structured documentation, and technical presentations.

[Design & Media Production: technical visual design, academic presentation design, and video editing/post-production for educational and engineering content.](#)

PERSONAL SKILLS

- Strong mentoring and instructional communication in classroom and laboratory environments.
- High responsibility, organization, and safety awareness in technical settings.
- Analytical thinking with disciplined validation and iterative improvement.
- Adaptability and continuous learning across multidisciplinary engineering topics.

LANGUAGES

Arabic (Native)
English (Intermediate)

TRAINING & COURSES

05/2026 **IEEE Authorship and Open Access Symposium: Tips and Best Practices to Get Published from IEEE Editors**

IEEE — Certificate of Participation, 90 minutes.

09/2021 **Training Course in Translation – Al-Kindi Institute for Languages & Computer, Yemen-Taiz**

60 Hours (Grade: Excellent – 98%)(issued 25/02/2021)

04/2025 **eSmarTA 2025 (IEEE) — Conference Participation**

Attendance certificate and multiple participation certificates (paper presentations).

05/2026 **Introduction in Artificial Intelligence**

Edraak — Level Completion Certificate, 3 hours.

05/2026 **Train-the-Trainer Program: The Novice Trainer Level**

Edraak — International Board of Certified Trainers (IBCT) — Level Completion Certificate

02/2024 **Internet of Things (IoT) — Edraak Platform**

Level Completion Certificate (issued 25/02/2024).

02/2024 **Influential Leadership & Effective Management— Edraak Platform**

Level Completion Certificate (issued 24/02/2024).

I also hold several certificates in English language and computer-related courses, which are available upon request. Furthermore, I am currently pursuing the International Computer Driving License (ICDL) and expect to complete it in the near future.

Industrial & Technical Training

08/2024 **Cooperative Industrial Training — General Industries & Packages Co. (GenPack), Taiz, Yemen**

- Completed cooperative industrial training within a packaging and industrial production environment.
- Gained practical exposure to production workflows, operational procedures, workplace discipline, and industrial safety practices.
- Observed engineering-related plant operations and strengthened understanding of industrial process organization and applied technical work.

- 08/2024–09/2024 **Industrial Training — National Dairy & Food Co. (Nadfood), Taiz, Yemen**
- Completed industrial training in the Engineering Department with exposure to plant utilities, production systems, and maintenance-related operations.
 - Trained on the working principles of generators, boilers, air compressors, sterilizers, and refrigeration systems.
 - Observed production-line operations across dairy, juice, yogurt, and food-processing plants.
 - Gained practical exposure to PLC-based production control compared with classical control approaches.
 - Identified and studied industrial pumps, valves, and motors used in production and utility systems.

- 01/2018 **Electronic Control Training — Yemen Company for Industry & Commerce (YCIC – HSA Group), Yemen**
- Completed practical training in electronic control systems within an industrial company environment.
 - Participated actively in applied technical activities related to industrial control practice.
 - Gained early hands-on exposure to control-system applications and industrial panel-level technical work.

AWARDS & RECOGNITION

- 2016/2017 **Academic Distinction — Ranked First in Cohort and Second Nationally Technical Industrial Institute, Sana'a, Yemen**
Awarded for outstanding performance in Industrial Control Systems / Electronics with a cumulative grade of 92.27% (Excellent).
- 2014–2016 **Academic Distinction — First Rank in Cohort Technical Industrial Institute, Al-Hasab, Taiz, Yemen.**
Recognized for excellent academic and practical performance during the first and second vocational levels in Industrial Control Systems / Electronics.
- 06/2026 **Certificate of Appreciation — Scientific Research and Academic Writing Program Leadership Al-Shaheeda Na'mah Rassam School, Taiz, Yemen**
Awarded in recognition of dedicated mentorship, training, and guidance in the Scientific Research and Academic Writing Program. The certificate acknowledges contributions to developing students' academic writing, critical reading, research engagement, scholarly confidence, and participation in publication-oriented academic activities.
- 2026 **Certificate of Recognition — Professional Excellence and Dedicated Service Al-Shaheeda Na'mah Rassam School, Taiz, Yemen**
Recognized for professional excellence, dedicated service, reliability, teamwork, problem-solving ability, and contributions to teaching, technical training, laboratory support, STEM education, and academic guidance.
- 2026 **Certificate of Achievement — Outstanding Contribution to Laboratory Development Al-Shaheeda Na'mah Rassam School, Taiz, Yemen**
Recognized for contributions to laboratory development, equipment organization, experiment preparation, technical troubleshooting, safety-aware laboratory practice, and hands-on STEM learning.

- 2022–2023 **Certificate of Appreciation — Physics Laboratory Equipment and Experiment Development**

Al-Ghad Al-Mushreq School, Taiz, Yemen

Recognized for contributing to the preparation and development of physics laboratory equipment, electronic experiments, and practical experimental setups supporting hands-on physics learning.

2023 Certificate of Appreciation — Student Recognition for Teaching Excellence

Al-Ghad Al-Mushreq School , Taiz, Yemen

Awarded by students in appreciation of dedicated teaching, supportive academic guidance, and positive contribution to the classroom learning environment.

LETTERS OF RECOMMENDATION

1. **Ass. Prof. Dr. Radhwan A. A. Saleh, Department of Software Engineering,** Faculty of Engineering, Kocaeli University, Türkiye — academic and research recommendation in robotics, computer vision, control systems, scholarly publishing, and graduation project supervision.
2. **Dr. Mubaarak A. A. Saif, Department of Energy Systems Engineering,** Faculty of Technology, Kocaeli University, Türkiye — research recommendation in power systems, power electronics, EV fast-charging systems, PV-integrated energy conversion, grid-interface control, and MATLAB/Simulink-based modeling and simulation.
3. **Ass. Prof. Dr. Mustafa Ghaleb, Department of Software Engineering,** Faculty of Engineering, Kocaeli University, Türkiye — research recommendation in cybersecurity, federated learning, explainable AI, trustworthy machine learning, systematic review methodology, and security digital twins.
4. **T. Rajaa Al-Dubai, Manager, Al-Shaheeda Na'mah Rassam School, Taiz, Yemen** — professional experience and institutional recommendation in STEM education, laboratory development, technical training, academic guidance, and student research mentorship.
5. **Eng. Abdumalek Ahmed Abdullah Fadhel, Technical Industrial Institute, Sana'a, Yemen** — academic recommendation in industrial control, electronics, academic discipline, and technical learning potential.
6. **Eng. Mohammad Ahmad Ahmad Saba, Technical Industrial Institute, Sana'a, Yemen** — academic recommendation in vocational technical education, motivation, discipline, and readiness for further academic study.

[Additional academic recommendation letters from the Technical Industrial Institute, Sana'a&Taiz, Yemen, are available upon request.](#)

COMMUNITY ENGAGEMENT & LEADERSHIP

2021 Founding Member — Sina'a Union, Taiz, Yemen

- Recognized as one of the founding members of Sina'a Union.
- Contributed to planning, coordination, and organizational activities supporting youth, educational, and community-oriented initiatives.

2021 Community Initiative Participation — Taiz Youth Festival, Taiz, Yemen

- Recognized for contributing to the success of the Taiz Youth Festival initiative.

- Participated in youth-oriented community activities aimed at promoting creativity, collaboration, and civic engagement.

2021 Training in Union Activities and Initiative Planning — Sina'a Union, Taiz, Yemen

- Completed training related to union activities, planning, and initiative development.
- Gained practical exposure to teamwork, organizational communication, leadership participation, and community-service planning.

PUBLICATIONS

PUBLICATIONS

1. Saleh, R. A., **Nasser, S. F.**, & Abdullah, T. S. (2025, August). Advances in Error Compensation for Robotic Manipulators: A Systematic Review from Geometric Calibration to AI-Driven Hybrid Control. In *2025 5th International Conference on Emerging Smart Technologies and Applications (eSmarTA)* (pp. 1-10). IEEE.
<https://doi.org/10.1109/eSmarTA66764.2025.11131700>
2. Saleh, R. A., **Nasser, S. F.**, Al Mashwali, M. M., Taha, E. M., & Al-Noor, O. A. (2025, August). A Survey on Camera-Based Measurement Systems: Techniques, Applications, and Challenges. In *2025 5th International Conference on Emerging Smart Technologies and Applications (eSmarTA)* (pp. 1-9). IEEE.
<https://doi.org/10.1109/eSmarTA66764.2025.11132174>
3. Saleh, R. A., **Nasser, S. F.**, Almakthay, A. F., & Al-Koshab, M. M. (2025, August). A Comprehensive Review of Transcranial Direct Current Stimulation (tDCS): Mechanisms, Cognitive Effects, Applications, and Future Directions. In *2025 5th International Conference on Emerging Smart Technologies and Applications (eSmarTA)* (pp. 1-8). IEEE.
<https://doi.org/10.1109/eSmarTA66764.2025.11132143>
4. Saleh, R. A., Alqudsi, Y., Al-Hajj, A. A., Abdullah, A. A., **Nasser, S. F.**, & Amrani, M. A. (2025, August). Advancements in Robotic Arm Technologies: Precision, Intelligence, and Cross-Sector Applications. In *2025 5th International Conference on Emerging Smart Technologies and Applications (eSmarTA)* (pp. 1-10). IEEE.
<https://doi.org/10.1109/eSmarTA66764.2025.11132178>

IN PROCESS PUBLICATIONS

MANUSCRIPTS SUBMITTED TO IEEE CONFERENCES — In process
Status: Accepted with revision required for oral presentation at the 6th International Conference of Smart Technology and Applications (eSmarTA 2026). Camera-ready versions and correction tables are under preparation.

1. Control-Oriented Design and Performance Evaluation of a Three-Terminal Storage-Assisted EV Fast-Charging System with an Interleaved Bidirectional Buck–Boost Converter **(Research paper)**
2. A Review of Electric Vehicle Charging Systems: Converter Architectures, Control Optimization, and PV-Integrated Energy Management **(Review paper)**
3. Toward Trust-by-Design in Federated Learning for Healthcare: A Taxonomic Review of Privacy, Explainability, and Accountability **(Review paper)**
4. Data-Driven Deep Learning for Imaging Through Scattering Media: Architectures, Learning Paradigms, and Methodological Frontiers **(Review paper)**

MANUSCRIPTS

RESEARCH PAPERS

Projects in progress: collaborative optimization refinement, algorithm preparation, and simulation-based result validation are in progress.

1. Hybrid Geometric–Sliding Mode Control of Quadrotor UAVs with Disturbance Rejection Optimized

2. Optimization-Assisted Model Predictive Control for Grid-Compliant PV-Integrated Bidirectional EV Fast-Charging Systems
3. Reliability-Constrained Multi-Objective Optimization of an Off-Grid PV–Wind–Diesel–Battery Hybrid Energy System: A HOMER-Benchmarked Cost–Renewability Assessment

REVIEW PAPERS

Review papers — completed manuscript; under collaborative review and pre-submission refinement.

1. Explainable and Privacy-Preserving Federated Learning in Healthcare: A Systematic Review and Taxonomic Synthesis
2. Security Digital Twins for SOC Engineering: Threat Simulation, Detection Content Generation, and Response Validation: A Systematic Review
3. Automating Cyber Security Penetration Testing Tasks with AI Agents using LLMs: A Systematic Review

RESEARCH OF INTERESTS

1. Advanced Artificial Intelligence, Machine and Deep Learning, and Image Processing.
2. Medical Image Analysis and Imaging Biomarkers.
3. LLMs, and Foundation Models.
4. Metaheuristic Optimization algorithms and their applications.
5. Pattern Recognition, Feature Selection Algorithms and Federated Learning.
6. Explainable Artificial Intelligence.

REFERENCES

1. Researches and graduation project supervisor Ass. Prof. Dr. Radhwan Ali Abdulghani Saleh, Assistant Professor, Department of Software Engineering, Kocaeli University, Türkiye:
radhwan.saleh@kocaeli.edu.tr
2. Research co-supervisor Dr. Mubaarak A. A. Saif, Department of Energy Systems Engineering, Faculty of Technology, Kocaeli University, Türkiye:
mubarak.saif@kocaeli.edu.tr
3. Research co-supervisor Ass. Prof.Dr. Mustafa Ghaleb, Assistant Professor, Department of Software Engineering, Faculty of Engineering, Kocaeli University, Türkiye:
Mustafa.ghaleb@kocaeli.edu.tr
4. Professional reference T. Rajaa Al-Dubai, Manager, Al-Shaheeda Na'mah Rassam School, Taiz, Yemen:
ne3marasam@ne3marasam.org